

Website Mockup Generator — PRD

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Overview

Paste a website link and get polished, photo-realistic device mockups of every important page in under two minutes, with no Photoshop and no manual screenshots.

Problem. Showing a website well today takes three manual jobs: taking clean screenshots of each page at desktop, tablet and mobile sizes; finding mockup photos; and warping each screenshot into each photo in Photoshop or Figma. For an agency presenting work to clients, building a portfolio or posting on social media, this often takes hours per site and the quality depends on who does it.

Product. A web tool that automates the whole chain:

1. **Discover** the site's pages from its sitemap or navigation.
2. **Capture** each chosen page at three device sizes with a headless browser, cleaning up cookie banners and pop-ups.
3. **Compose** the screenshots into real mockup photos by perspective-warping each one into the screen area defined by four corner points.
4. **Export** high-resolution images, social-size presets, or a ZIP of the whole set.

Edge over existing tools. Shots.so, Screely, Smartmockups and Previewed require the user to upload screenshots by hand. This tool starts from a link, handles many pages at once, and supports Arabic and RTL sites properly.

Goals, non-goals and success metrics

The product succeeds when a non-designer turns a URL into a client-ready mockup set without opening another tool.

Goals

- Turn one URL into a set of mockups for multiple pages in a single flow.
- Produce output realistic enough to use in client decks, portfolios and social posts without retouching.
- Make adding new mockup photos fast (about one minute each) so the library grows easily.
- Render Arabic and RTL sites correctly.

Non-goals (v1)

- Editing the website's content before capture.
- Capturing pages behind a login automatically (manual upload covers this).
- AI-generated mockup scenes.
- A full design editor like Figma or Canva.

Success metrics (proposed targets)

Metric	Target
Time from URL to first downloaded mockup	under 2 minutes
Captures that need no recapture or upload fallback	85% or more
Mockups used without manual retouching	90% or more
Time to add a new mockup photo to the library	about 1 minute
Pages per project, on average	3 or more

Users and use cases

The first users are the BeCorp team; the same needs apply to freelancers and other agencies if the tool is later sold.

User	What they need	Typical use
Account manager / strategist	Fast visuals of a client's current or new site	Pitch decks, audit reports, proposals
Designer	High-quality, consistent mockup sets	Portfolio, Behance, case studies
Social media manager	Ready-sized images	Launch posts for Instagram, LinkedIn, Facebook
Developer	Proof of delivery across devices	Handover to the client, showing responsive behaviour
Admin (internal)	A way to grow and curate the mockup library	Uploading photos and marking screen corners

Key use cases

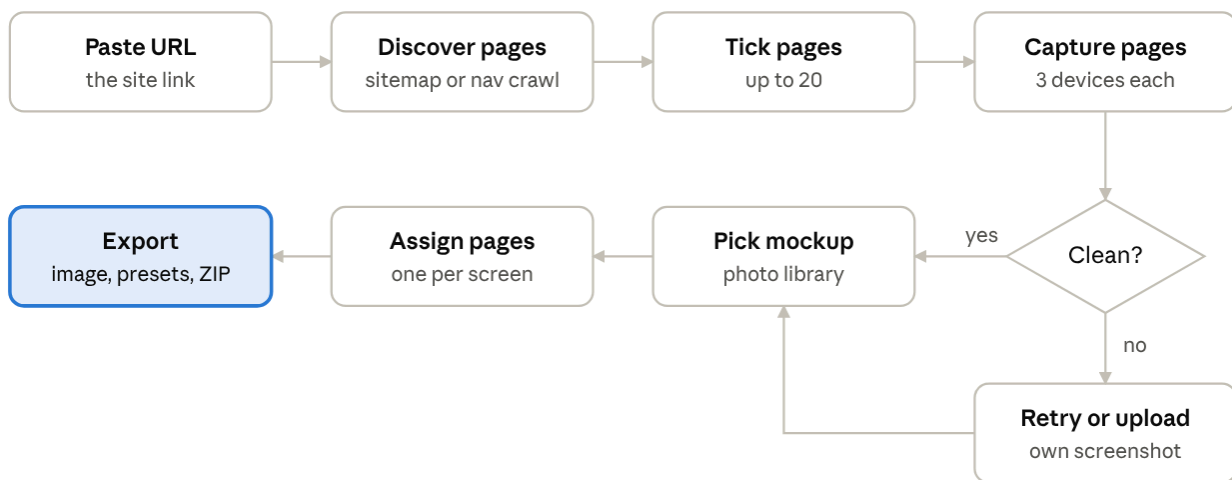
1. **Client launch post:** a new site goes live; the team captures Home, Services and Contact and posts a three-device scene the same day.

2. **Portfolio set:** one mockup style applied to every page of a project for a consistent Behance gallery.
3. **Audit or pitch:** a prospect's current site shown in mockups next to the proposed redesign.
4. **Arabic site showcase:** an RTL site captured with correct fonts and layout.

User flow

The whole journey is eight steps, and a failed capture never stops it: the user retries or uploads their own screenshot and carries on.

One link in, a full mockup set out



user flow · 8 steps, 1 fallback

Multi-page and multi-screen work happens in Assign pages: each screen in the chosen photo gets one page, and batch mode repeats the mockup for every page.

Page discovery

The tool finds the site's pages automatically and lets the user tick which ones to capture.

ID	Requirement	Priority
PD-1	Accept a URL; normalise it (add https, strip tracking parameters) and confirm it loads.	Must
PD-2	Read <code>robots.txt</code> and <code>sitemap.xml</code> (including sitemap index files) to list pages.	Must

ID	Requirement	Priority
PD-3	If there is no sitemap, crawl links in the header nav, footer and homepage body, same domain only, up to 2 levels deep.	Must
PD-4	Cap discovery at 200 URLs and 30 seconds; show what was found so far if the cap is hit.	Must
PD-5	Show pages as a checklist with title, path and a small favicon; Home is pre-selected.	Must
PD-6	Group repeated templates (e.g. <code>/product/*</code> , <code>/blog/*</code>) and show one sample per group, expandable.	Should
PD-7	Let the user add URLs manually and search or filter the list.	Must
PD-8	Detect language variants (<code>/ar/</code> , <code>/en/</code> , <code>hreflang</code>) and let the user pick one or both.	Should
PD-9	Limit the selection to 20 pages per project in v1.	Must

Capture engine

Each selected page is loaded in headless Chromium at three device sizes, cleaned up, and saved as a sharp 2x image.

Viewports (defaults, editable per project)

Device	Width × height (CSS px)	Scale	Output size
Desktop	1440 × 900	2x	2880 × 1800
Tablet	834 × 1194	2x	1668 × 2388
Mobile	390 × 844	3x	1170 × 2532

Requirements

ID	Requirement	Priority
CE-1	Capture modes: above-the-fold (viewport only) and full-page (whole scroll height, capped at 15,000 px).	Must

ID	Requirement	Priority
CE-2	Wait for network idle, web fonts (<code>document.fonts.ready</code>) and images before shooting.	Must
CE-3	Scroll the page top to bottom once to trigger lazy-loaded images and scroll animations, then return to top.	Must
CE-4	Hide cookie banners, chat widgets and pop-ups using a maintained blocklist of selectors plus common consent-tool handling.	Must
CE-5	Freeze animations and carousels (inject CSS that sets animation and transition durations to zero).	Must
CE-6	Fix sticky headers in full-page mode so they appear once at the top, not repeated.	Should
CE-7	Optional: dark-mode capture (<code>prefers-color-scheme: dark</code>), hide specific elements by CSS selector, add a delay.	Should
CE-8	Arabic and RTL: install Noto Sans Arabic, Cairo, Tajawal and IBM Plex Sans Arabic on workers; never fall back to boxes.	Must
CE-9	Timeout of 45 s per page; on failure, show the reason and offer Retry or Upload my own screenshot.	Must
CE-10	Run up to 3 pages in parallel per project; queue the rest; show per-page progress.	Must
CE-11	Results gallery: pages as rows, devices as columns; recapture any single cell.	Must
CE-12	Cache captures for 24 hours by URL, viewport and options to avoid duplicate work.	Could

Mockup library and corner picker

Every mockup is a real photo plus the exact position of each screen in it, marked once by an admin and reused for every render.

Library (user side)

ID	Requirement	Priority
ML-1	Grid of mockup photos with filters: device type (desktop, laptop, tablet, mobile, multi-device), scene (desk, hand, studio, outdoor), colour tone, orientation.	Must
ML-2	Each card shows a live preview using the user's own Home capture, not a placeholder.	Should
ML-3	Favourites, and a "recently used" row.	Could

Admin corner picker

ID	Requirement	Priority
CP-1	Upload a photo (JPG/PNG, at least 3000 px on the long side).	Must
CP-2	Add one or more screens; for each, click the four corners in order top-left, top-right, bottom-right, bottom-left.	Must
CP-3	Drag handles with a magnifier loupe for pixel-accurate placement; show a live warped test image while adjusting.	Must
CP-4	Per screen: device type (sets which viewport it receives), screen corner radius, and an optional mask PNG for notches or curved edges.	Must
CP-5	Optional overlay PNG per mockup (glass reflection, a thumb over the screen) drawn above the screenshot.	Should
CP-6	Optional shadow/lighting blend: multiply the screen layer with a light map so it matches the photo's lighting.	Could
CP-7	Metadata: title, tags, licence source and licence type, required attribution. A mockup cannot be published without licence info.	Must
CP-8	Draft / published status; only published mockups appear to users.	Must

Stored screen definition (example)

```
{
  "screenId": "s1",
  "device": "mobile",
  "corners": {"tl": [1204, 388], "tr": [1712, 402], "br": [1690, 1466], "bl": [1182, 1450]},
  "cornerRadius": 38,
  "maskUrl": "masks/iphone-notch.png",
  "zIndex": 1
}
```

Photo sources. Free-licence photos (Unsplash, Pexels; check each licence), paid packs (Freepik, Envato), or photos shot in-house with BeCorp devices, which gives a unique look and no licence risk. Avoid visible brand logos if the tool will be sold.

Composition and rendering

Each screenshot is cropped to its screen's shape, perspective-warped onto the four corners, and layered between the photo and its overlay.

Render steps per screen

1. Pick the capture matching the screen's device type (mobile screen → mobile capture).
2. Crop the capture to the screen's aspect ratio, measured from its corner points; crop from the top by default, never stretch.
3. Compute the homography from the crop's rectangle to the four corners and warp with bicubic interpolation (OpenCV `getPerspectiveTransform` + `warpPerspective`).
4. Apply the rounded-corner radius and the mask, with 1 px anti-aliased edges.
5. Composite: base photo → warped screens (by z-index) → optional light map (multiply) → overlay PNG.
6. Render at the photo's native resolution, then produce export sizes.

ID	Requirement	Priority
CR-1	Perspective warp with no visible jagged edges or seams at 100% zoom.	Must
CR-2	Multi-screen mockups: the user assigns a page to each screen (e.g. Home on the laptop, Contact on the phone); sensible defaults auto-fill.	Must
CR-3	Scroll offset per screen: choose which part of a full-page capture shows (drag up or down in the preview).	Should

ID	Requirement	Priority
CR-4	Batch mode: apply one mockup to every captured page and render the whole set.	Must
CR-5	Presentation layouts without photos: grid collage of pages, and a tall frame showing a full-page scroll.	Should
CR-6	Preview render in under 3 s at reduced size; final render in under 10 s per image.	Must
CR-7	Same inputs always give the same output (deterministic), so re-exports match.	Must

Customisation and export

Users get a few high-impact controls and one-click exports, not a full design editor.

Customisation

ID	Requirement	Priority
CX-1	Swap the page shown on each screen.	Must
CX-2	For photo-free layouts: background (solid, gradient, colours pulled from the site's brand), padding, shadow strength.	Should
CX-3	Auto-extract the site's main colours from its CSS and logo to suggest backgrounds.	Could
CX-4	Optional text: headline and client logo placed on presentation layouts.	Could

Export

ID	Requirement	Priority
EX-1	PNG and WebP at native resolution; JPG at 90% quality.	Must
EX-2	Size presets: Instagram post 1080 × 1350, Instagram story 1080 × 1920, LinkedIn 1200 × 627, Behance 1400 wide, slide 1920 × 1080, 4K 3840 × 2160.	Must

ID	Requirement	Priority
EX-3	Download one image, or a ZIP of the whole set named <code>site_page_device_mockup.png</code> .	Must
EX-4	Shareable gallery link for a project (read-only, can be revoked).	Should
EX-5	Scrolling MP4 of a full-page capture inside a device frame (10 to 20 s).	Could (v2)
EX-6	Attribution text included in the ZIP when a mockup's licence requires it.	Must

Data model

A project holds pages, each page holds one capture per device, and a render joins captures to a mockup's screens.

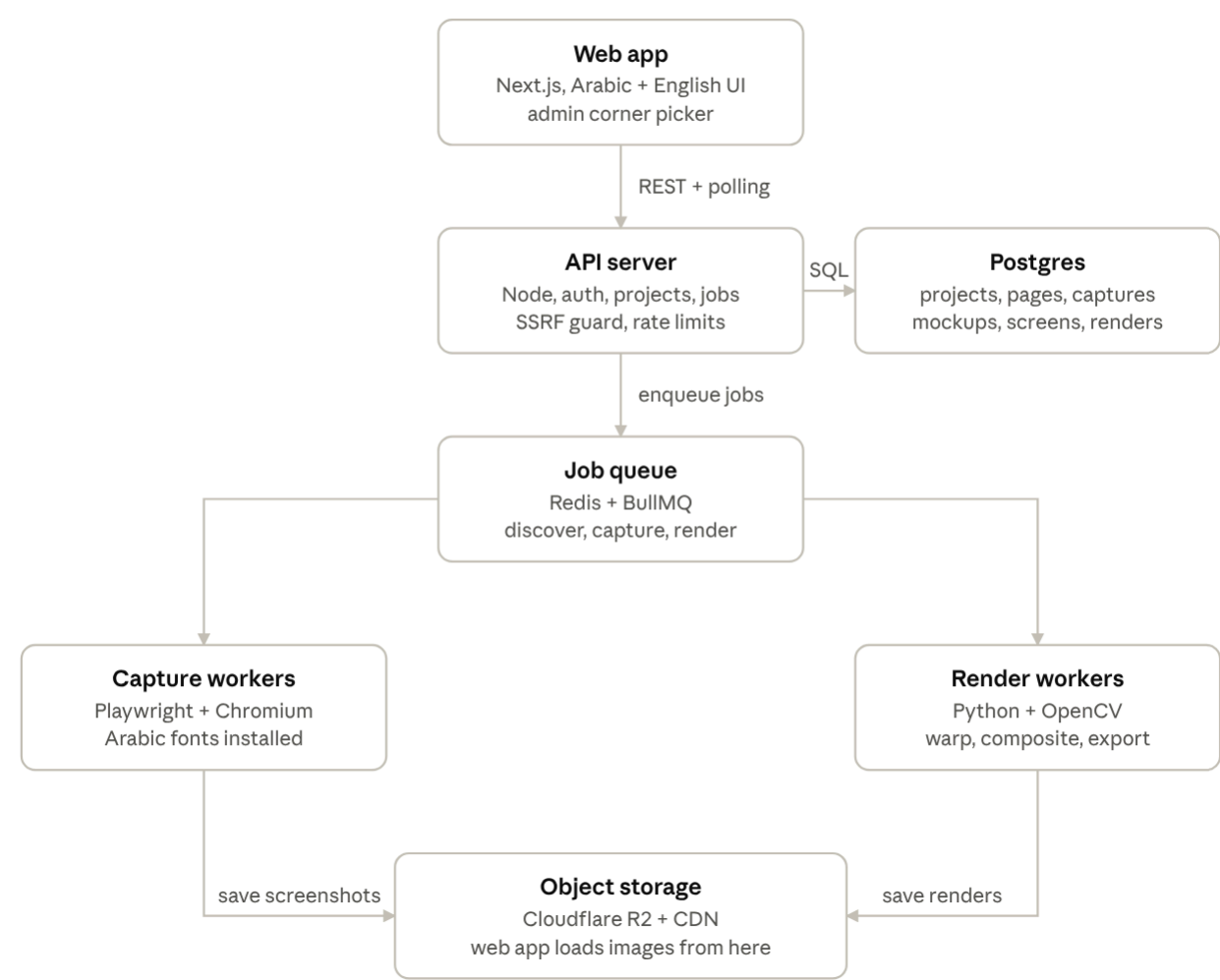
Entity	Key fields	Relations
User	id, name, email, role (admin, member), locale	owns Projects
Workspace	id, name, plan, usage counters	has Users, Projects
Project	id, rootUrl, title, language, createdAt	has Pages, Renders
Page	id, url, title, templateGroup, order, selected	belongs to Project; has Captures
Capture	id, device, mode (fold, full), width, height, imageUrl, status (queued, running, done, failed), error, source (auto, upload), options JSON	belongs to Page
Mockup	id, title, photoUrl, width, height, tags, scene, licenceSource, licenceType, attribution, overlayUrl, lightMapUrl, status (draft, published)	has Screens
Screen	id, device, corners JSON, cornerRadius, maskUrl, zIndex	belongs to Mockup
Render	id, mockupId, assignments JSON (screenId → captureId + scrollOffset), outputs JSON (size → fileUrl), status	belongs to Project

Entity	Key fields	Relations
Job	id, type (discover, capture, render, export), payload, attempts, status, timings	queue record

Architecture and tech stack

The web app only takes requests and shows results; discovery, capture and rendering run as background jobs on separate workers that scale on their own.

Heavy work runs in workers behind a queue, never in the web app



system architecture · 7 components

Workers update job status in Postgres; the web app polls it to show per-page progress.

Layer	Choice	Why
Front end	Next.js (React) + Tailwind, RTL support	Fast to build, one codebase for app and admin
Corner picker	HTML canvas + <code>matrix3d</code> live preview	Instant feedback while dragging handles
API	Next.js API routes or a small Node (Fastify) service	Same language as the front end
Database	Postgres (Supabase or self-hosted)	Relational data, JSON fields for corners and options
Queue	Redis + BullMQ	Retries, concurrency limits, progress events
Capture	Playwright with Chromium, in Docker	Reliable screenshots, device emulation, full-page mode
Rendering	Python + OpenCV (<code>warpPerspective</code>) + Pillow	Best-quality warp and compositing
Storage	Cloudflare R2 or Backblaze B2 + CDN	Cheap egress for large images
Hosting	VPS with Docker Compose for v1 (e.g. Hetzner), workers as separate containers	Low cost, easy to add workers

Non-functional requirements

The tool must be fast enough to use live in a meeting, safe to point at any URL, and correct for Arabic.

Area	Requirement
Performance	Discovery under 30 s; one page × 3 devices under 30 s; 10 pages under 3 minutes; preview render under 3 s.
Scale (v1)	10 concurrent projects, 3 browser workers, horizontal scaling by adding workers.
Security: SSRF	Block private and internal IPs (10.x, 192.168.x, 127.x, 169.254.x, localhost) after DNS resolution and on every redirect. Only http/https.

Area	Requirement
Security: isolation	Each capture runs in a fresh browser context; no downloads, no file access, JavaScript dialogs auto-dismissed, 45 s hard kill.
Abuse	Rate limit per user (e.g. 50 pages/hour); log every captured domain.
Storage	Captures and renders on S3-compatible storage (Cloudflare R2 or Backblaze B2); signed URLs; captures deleted after 30 days unless saved.
Privacy	Captures of public pages only in v1; no cookies or credentials stored.
Localisation	Interface in Arabic and English with full RTL layout; Arabic fonts on capture workers (CE-8).
Licensing	Every published mockup has licence source and type recorded; attribution exported when required.
Reliability	Failed jobs retried twice with backoff; a failed page never blocks the rest of the project.
Browser support	Latest Chrome, Edge, Safari, Firefox; usable on tablet; view-and-download on mobile.

Risks and mitigations

The biggest risk is capture quality on real-world sites; every capture risk has a manual upload fallback.

Risk	Impact	Mitigation
Sites block headless browsers (Cloudflare, bot checks)	Capture fails	Realistic user agent and headers, retry once; then Upload my own screenshot.
Cookie banners and pop-ups slip through	Ugly output	Maintained selector blocklist; "hide element" picker in the gallery; easy recapture.

Risk	Impact	Mitigation
Lazy images or animations caught mid-load	Blank or half-faded sections	Scroll pass, network-idle wait, animation freeze, optional extra delay.
Pages behind a login	Cannot capture	Out of scope for v1; manual upload. v2: user-supplied session cookie.
Arabic text renders as boxes or wrong font	Unusable for Iraqi clients	Fonts pre-installed on workers; Arabic test site in the QA suite.
Warp looks pasted on	Low perceived quality	Overlay and light-map layers; only publish mockups that pass visual review.
Mockup photo licence problems	Legal risk if sold	Licence fields required before publishing; prefer in-house photos.
Using the tool to scan internal networks (SSRF)	Security incident	IP blocklist after DNS and on redirects; isolated workers.
Server cost of headless Chromium	Slow or expensive at scale	Concurrency limits, caching, workers scaled by queue length.

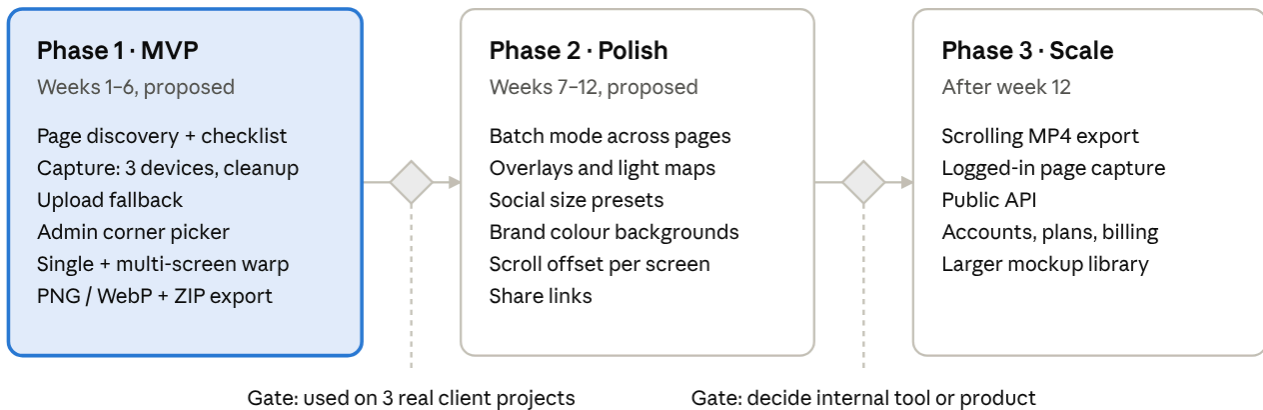
MVP scope and roadmap

The MVP is the shortest path from a URL to a real photo mockup of several pages; everything else waits until the team has used it on real client work.

In the MVP: URL input, page discovery and checklist, capture at 3 devices with cleanup, upload fallback, admin corner picker, 10 to 15 in-house mockup photos, single and multi-screen warp, PNG/WebP and ZIP export, Arabic and English interface.

Not in the MVP: batch mode, overlays and light maps, social presets, brand colour extraction, share links, video, logged-in pages, accounts and billing.

Ship a usable MVP first, then polish, then scale



roadmap · 3 phases, 2 gates

Gate 1 checks the output is good enough for client work; gate 2 decides whether accounts, billing and a public launch are worth building.

Open questions

- ☐ Internal BeCorp tool only, or a product sold to other agencies and freelancers?
- ☐ Product name and brand.
- ☐ Shoot mockup photos in-house, buy packs, or both? How many at launch?
- ☐ Hosting: local server in Iraq or an overseas VPS (affects capture speed for Iraqi sites)?
- ☐ Should captures support Arabic and English versions of the same site side by side?
- ☐ Who on the team builds it, and is the 12-week estimate realistic for that team?